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Mandelbrot Upic

A Commodore 64 Ultimate demo that generates a Mandelbrot fractal on-device at 64 MHz turbo, packs it directly into Upic format (a 16-color, 384x256 border-color raster picture technique), and displays it live.

Spun off from [landofficeandfire](#), which built and hardware-validated the Upic viewer this project reuses. See CREDITS.md for full attribution and docs/ARCHITECTURE.md/ docs/MANDELBROT_ALGORITHM.md for how it works and what's planned.

Status: fixed-point escape-time generator implemented and confirmed working on real hardware, live-updating the picture as it renders. Interactive zoom (pick a region of the current picture, regenerate at that scale) is implemented on the zoom- feature branch -- not yet confirmed on real hardware, see its own section below for current caveats. See docs/MANDELBROT_ALGORITHM.md for the full design.

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Controls

Generation starts automatically on launch and the picture builds up live. Once it completes, 4 corner-marker sprites appear over the picture, outlining a zoom-target rectangle:

Input	Action
W/A/S/D	Move the active corner
Cursor keys (hold Shift for up/left)	Same, alternative for muscle memory
Joystick, port 2	Same, directional + fire
RETURN or joystick fire	Confirm the active corner, then the other
Q	Quit to BASIC

Picking a zoom target is a two-step process: move the first corner into place and confirm, then move the second (opposite) corner and confirm again -- this regenerates the fractal at the newly selected region, and the same corner-selection screen appears again once it completes, so zooming can be repeated. The selection must span at least 16 cells in both directions; confirming a smaller one is simply ignored so adjustment can continue.

Joystick **port 2** (\$DC00) is used deliberately, not port 1 -- port 1 shares hardware lines with the keyboard matrix and gives unreliable readings while keys are also being read, which happens every frame on this screen.

Zoom precision limit: the fractal coordinates use a fixed-point format with a finite number of fractional bits, which caps how far repeated zooming can go (roughly 16x total from the initial view) before individual pixel steps round down to zero and the picture stops changing with further zoom. This is a hard limit of the current implementation, not a bug.

Not yet confirmed on real hardware (zoom-feature branch, as of 2026-09-10): the corner sprites' on-screen position relative to the picture is a best-effort calibration guess (see ZOOM_SPRITE_X0/ZOOM_SPRITE_Y0 in include/zoom.c) -- if the markers don't visually line up with the picture, those two constants are the first (and should be the only) thing to adjust.

Building from source

Prerequisites

Tool	Purpose	Install
Oscar64	C cross-compiler targeting 6502/C64	build from source, see their README
wput	FTP deploy to the Ultimate device	<code>sudo apt install wput</code>
pandoc + texlive-xetex	Generate README.pdf (optional)	<code>sudo apt install pandoc texlive-xetex</code>

Deploy setup

Copy `.env.example` to `.env` and set `ULTIP1` to your Ultimate device's IP address:

```
cp .env.example .env
```

Make targets

Target	Effect
<code>make /make all</code>	Compile to <code>build/mandelupic.prg</code> , regenerate <code>README.pdf</code> , build the release ZIP
<code>make deploy</code>	FTP the compiled <code>.prg</code> to the Ultimate device set in <code>.env</code>
<code>make docs</code>	Regenerate <code>README.pdf</code> via pandoc
<code>make clean</code>	Remove build outputs

Requires **Ultimate firmware 3.15 or newer** (UCI auto-enable, palette control commands -- see UCILIBMANUAL.md).